

CS 507: Computer Architecture

Course Code	CS 507
Course Name	Computer Architecture
L-T-P-C	4-0-0-4
Prerequisites	CS201 Computer Organization or Equivalent
Intended for	BTech Computer Science Engineering (CSE) and Electrical Engineering (EE), MS, M. Tech. & PhD.
Distribution	Elective for Third and Final year B. Tech (CSE/EE), MS, M. Tech. in VLSI/Signal Processing and Communication & PhD
One-time approval	14th BoA (Sep 2016); 12th Senate (Mar 2017)
Revised	18th BoA (Sep 2017) - syllabus revised; 14th Senate (Oct 2017) - renamed from "Advanced Computer Architecture"

Course Contents

- **Introduction:** Defining Computer Architecture, Flynn's Classification of Computers, Metrics for Performance Measurement [4 Lectures]
- **Memory Hierarchy:** Introduction, Advanced Optimizations of Cache Performance, Memory Technology and Optimizations, Virtual Memory and Virtual Machines, The Design of Memory Hierarchy, Introduction to Pin Instrumentation and Cachegrind, Case Study: Memory Hierarchies in Intel Core i7 and ARM Cortex-A8. [8 Lectures]
- **Instruction Level Parallelism:** Instruction-level Parallelism: Concepts and Challenges, Basic Compiler Techniques for Exposing ILP, Reducing Branch Costs with Advanced Branch Prediction, Dynamic Scheduling, Advanced Techniques for Instruction Delivery and Speculation, Limitations of ILP, Multithreading: Exploiting Thread-Level Parallelism to Improve Uniprocessor Throughput, Modeling Branch Predictors using Pin Tool, Case Study: Dynamic Scheduling in Intel Core i7 and ARM Cortex-A8. [10 Lectures]
- **Thread Level Parallelism:** Introduction, Shared-Memory Multicore Systems, Performance Metrics for Shared-Memory Multicore Systems, Cache Coherence Protocols, Synchronization, Memory Consistency, Multithreaded Programming using OpenMP, Case Study: Intel Skylake and IBM Power. [10 Lectures]
- **Data Level Parallelism:** Introduction, Vector Architecture, SIMD Instruction Set Extensions for Multimedia, Graphics Processing Units, GPU Memory Hierarchy, Detecting and Enhancing Loop-Level Parallelism, CUDA Programming, Case Study: Nvidia Maxwell. [10 Lectures]

Computer Architecture Lab

- The lab experiments (assignments) will be designed to assist the theory covered in the class.

Text Books

1. J.L. Hennessy and D.A. Patterson. **Computer Architecture: A Quantitative Approach**, 5th Edition, Morgan Kaufmann Publishers, 2012.

References

1. J.P. Shen and M.H. Lipasti, **Modern Processor Design: Fundamentals of Superscalar Processors**, McGraw-Hill Publishers, 2005.
2. D.B. Kirk and W.W. Hwu, **Programming Massively Parallel Processors**, 2nd Edition, Morgan Kaufmann Publishers, 2012.
3. Pin - A Dynamic Binary Instrumentation Tool. <https://software.intel.com/en-us/articles/pin-a-dynamic-binary-instrumentation-tool>
4. Cachegrind: A Cache and Branch-Prediction Profiler. <http://valgrind.org/docs/manual/cg-manual.html>
5. OpenMP.
6. CUDA. <https://developer.nvidia.com/cuda-zone>